

Task 2 — Virtual Machine Setup on AWS EC2

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1. Objective

Launch a virtual machine on the AWS Free Plan, configure its firewall, connect to it over SSH, install the **nginx** web server, and verify it serves a live page — documenting each step with screenshots.

2. Setup summary

Item	Value
Instance name / ID	codeorbit-vm / i-0bfdff29c09534c84
Instance type	t3.micro (2 vCPU, Free Plan eligible)
Operating system	Amazon Linux 2023 (ami-01e082ac2f79f3918)
Region / Availability Zone	eu-north-1 (Stockholm) / eu-north-1b
Public IPv4 address	13.49.230.117
Storage	8 GiB gp3 root volume
Key pair	codeorbit-key (.pem, downloaded at launch)
Security group	launch-wizard-1 : SSH (22) from My IP + EC2 Instance Connect prefix list; HTTP (80) from 0.0.0.0/0
Application installed	nginx 1.30.5, enabled at boot, verified active

3. Steps performed

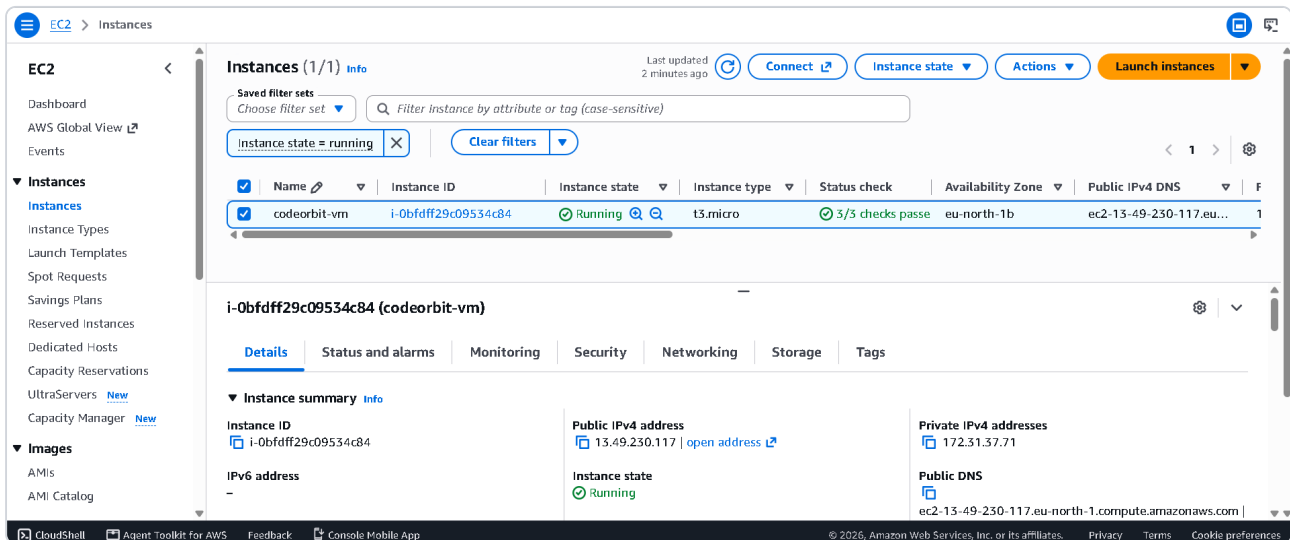
- Launched the instance** with the EC2 launch wizard: named it `codeorbit-vm`, kept the Amazon Linux 2023 AMI and `t3.micro` type, created the `codeorbit-key` key pair, enabled auto-assign public IP, created security group `launch-wizard-1` (SSH from My IP, HTTP from anywhere), and kept the default 8 GiB volume.
- Waited for 3/3 status checks** to pass before connecting.
- Connected over SSH** from Windows PowerShell using the downloaded `.pem` key (after fixing its Windows file permissions with `icacls` — see §5).
- Installed nginx:**

```
sudo dnf update -y
sudo dnf install -y nginx
sudo systemctl start nginx
sudo systemctl enable nginx
systemctl is-active nginx    # → active
```

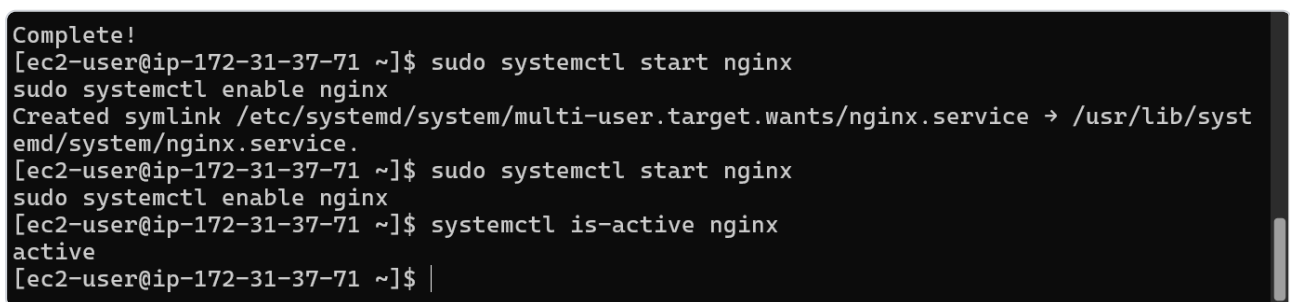
- Verified in the browser:** opened `http://13.49.230.117` — the "Welcome to nginx!" page loaded, proving the security-group HTTP rule and the server both work.

2. **Stopped the instance** after the screenshots to avoid burning Free Plan credits (a running VM costs a few cents/hour, mostly the public IPv4 address).

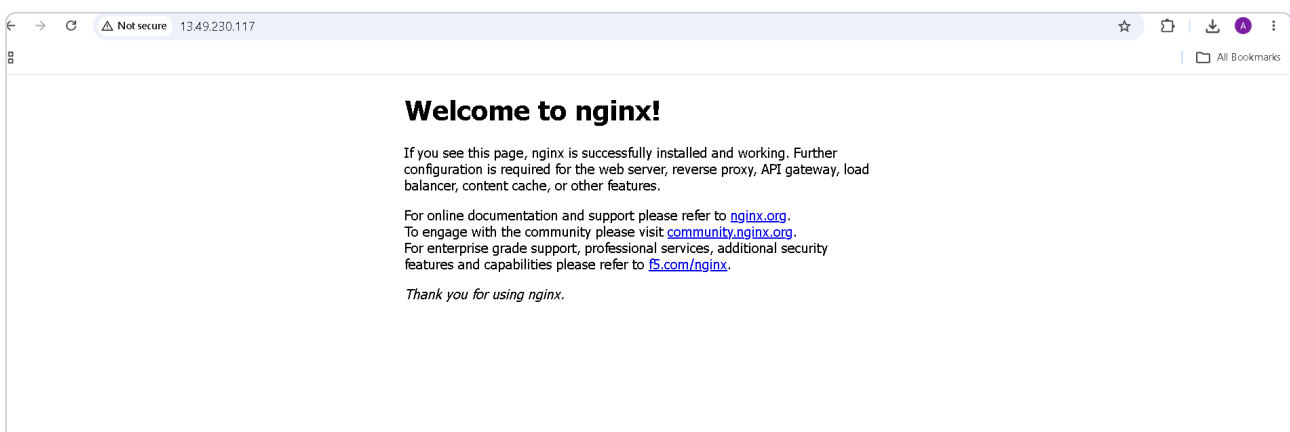
4. Screenshots



Screenshot 1 — EC2 Instances list: codeorbit-vm Running, 3/3 checks passed, public IP 13.49.230.117.



Screenshot 2 — SSH terminal: nginx installed and systemctl reports active.



Screenshot 3 — Browser at http://13.49.230.117 showing the nginx welcome page.

5. Issues faced and how they were fixed

- **EC2 Instance Connect failed at first.** Two causes: the status checks were still initializing, and I had picked the IPv6 EC2 Instance Connect prefix list while the instance has no IPv6 address. Fixed by selecting `com.amazonaws.eu-north-1.ec2-instance-connect` (IPv4) in the security group and waiting for 3/3 checks.

- **Windows SSH refused the .pem key** ("bad permissions / UNPROTECTED PRIVATE KEY FILE"). Fixed with `icacls codeorbit-key.pem /reset`, then `/grant:r` for the current user and `/inheritance:r` — after which `ssh -i codeorbit-key.pem ec2-user@13.49.230.117` connected on the first try.

6. Cost note

This build ran on the AWS Free Plan (\$100 in credits at signup, up to \$200 total, 6 months — no charges possible without upgrading to Paid). The few hours of `t3.micro` used here cost a few cents of credits; the instance was stopped immediately after the task.

7. Conclusion

The virtual machine was provisioned, secured, connected to, and turned into a working web server in under an hour, all on free-plan credits. The exercise demonstrated the full IaaS loop from Task 1's report: renting raw infrastructure (EC2), configuring the network boundary (security groups), and managing the software stack (nginx) myself.